

PolyTile 4.0:

Self-healing Ceramic Tiles

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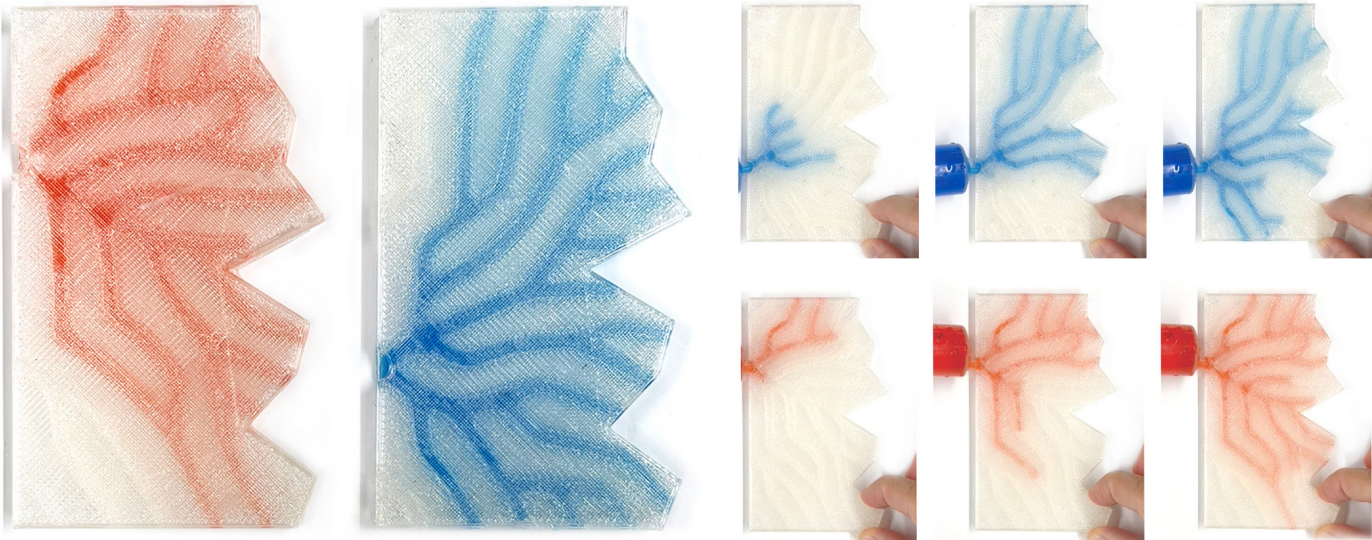
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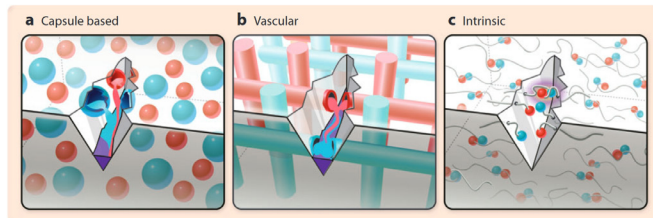
ABSTRACT

PolyTile 4.0 is a transdisciplinary research project merging architecture and bio-engineering to explore the self-healing potential of DNA hydrogel in ceramic materials. This project has developed a computational method using the Dijkstra Algorithm to create efficient storage channels within ceramic tiles for storing crack-healing agents. Fluid flow tests have validated its feasibility, establishing a method to connect multiple tiles and enabling healing agents to flow throughout the system. While self-healing tests on actual ceramic tiles are ongoing, the research aims to extend to concrete and other architectural materials, showcasing the potential of programmable materials for healing buildings and infrastructure.

1 Cracked Tiles with Optimized Channels for Storing Healing Agents and Fluid Flow Testing

SELF-HEALING MATERIAL

Self-healing materials mimic biological systems like human skin or bones, autonomously repairing damage caused by environmental factors, mechanical stress, or accidents. This innovation has been applied in many fields such as concrete (Sangadji and Schlangen, 2013) and polymers (Toohey et al., 2007). Self-healing systems are typically classified into three types: capsule-based, vascular, and intrinsic (Fig. 2)(Blaiszik et al., 2010). Capsule-based systems release healing agents upon damage but offer only single-use functionality, making them less sustainable. Intrinsic systems, relying on reversible chemical reactions within the material itself to trigger healing, depend heavily on advances in material science. Vascular systems, which feature capillary networks that store and distribute

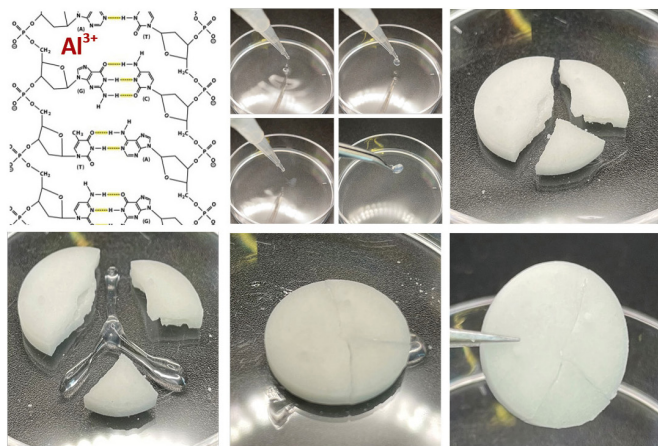


2 Three Types of Self-healing Approaches

healing agents, are more promising for architectural applications as they allow multiple repairs through external refilling. While most research has focused on suitable healing agents, less attention has been given to optimizing the storage and distribution networks. Our research addresses this gap by proposing a computational method to improve vascular system efficiency, enhancing material performance and sustainability.

DNA HYDROGEL AS A HEALING AGENT

When DNA solution meets aluminum chloride solution, a gelling process occurs, forming a solid compound with mechanical strength once dried (Fig. 3). By storing these

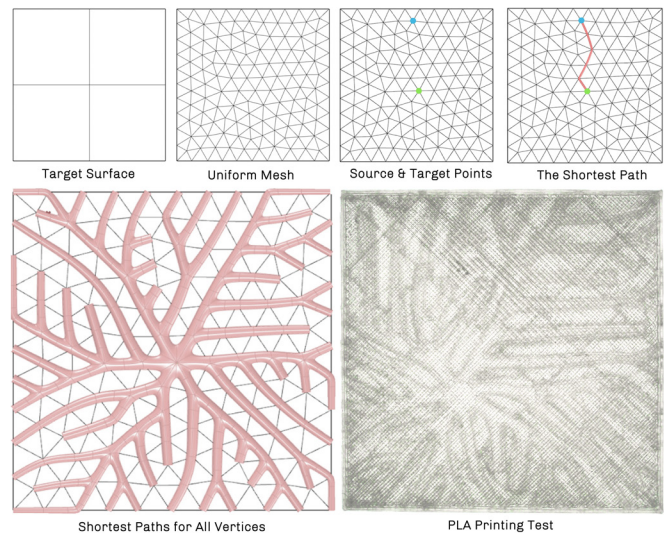


3 Applying DNA Hydrogel to Heal Cracks in Ceramic Disks

two healing agents inside ceramic tiles, self-healing functions can be achieved.

EFFICIENT STORAGE SYSTEM

The Dijkstra's algorithm was implemented using C# scripting on the Rhino Grasshopper platform to minimize material usage and production costs. This shortest-path finding algorithm generates unique branching patterns from mesh vertices to the healing agent source by finding the shortest path for each vertex. Converting these patterns into a channel system results in efficient storage, using minimal healing agents to cover the entire tile (Fig. 4).



4 An Efficient Storage Channel System Generated by the Dijkstra Algorithm

OPTIMIZED CHANNEL SYSTEM

Using this workflow, a PLA tile with internal channels was printed for initial fluid flow tests. The red dye indicated dead ends in the channel system, obstructing uniform liquid flow (Fig. 5). To address this, the algorithm used for pathfinding was researched in more detail. The algorithm can optimize for either total length or total step numbers. We discovered that using the total length criterion resulted in fewer dead ends. Then by connecting these dead ends to other channels, the geometry for printing was finalized. This new iteration enabled a smoother fluid flow (Fig. 6).

CRACK SIMULATION

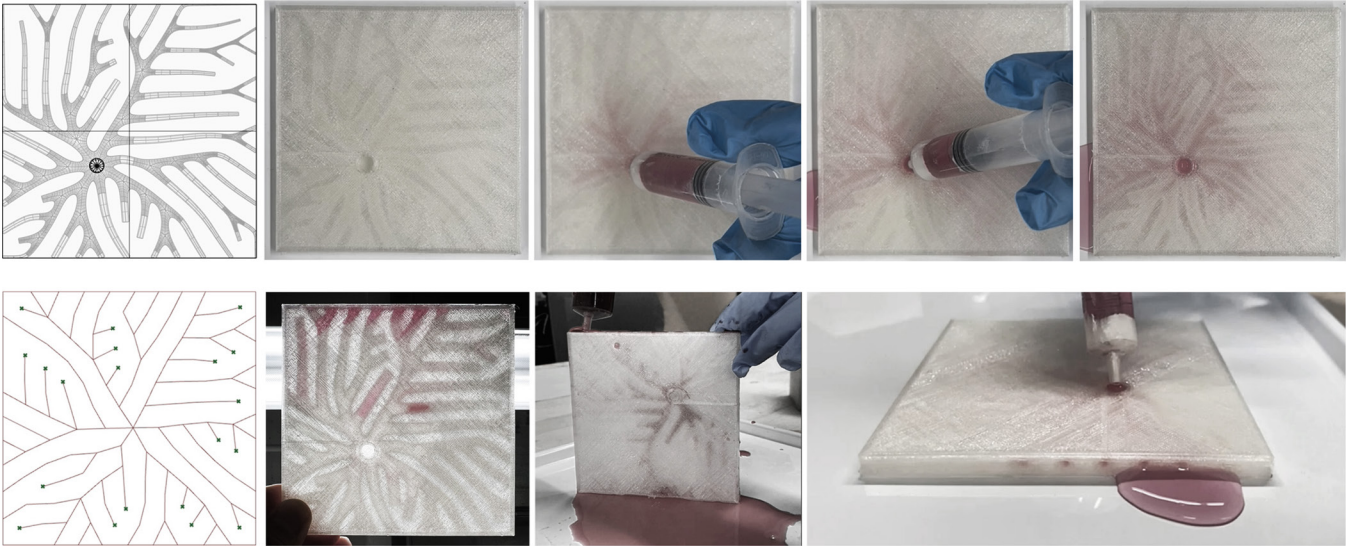
Two channel systems, isolated by a glaze layer, were designed to store the two healing agents separately. Upon cracking, water rehydrates the gel in both layers, causing them to flow out, react, and form a strong bonding structure to repair the crack. Fluid flow tests on PLA-printed tiles simulated this scenario with smooth and even liquid flow (Fig. 1 & Fig. 7).

MULTIPLE TILE CONNECTION

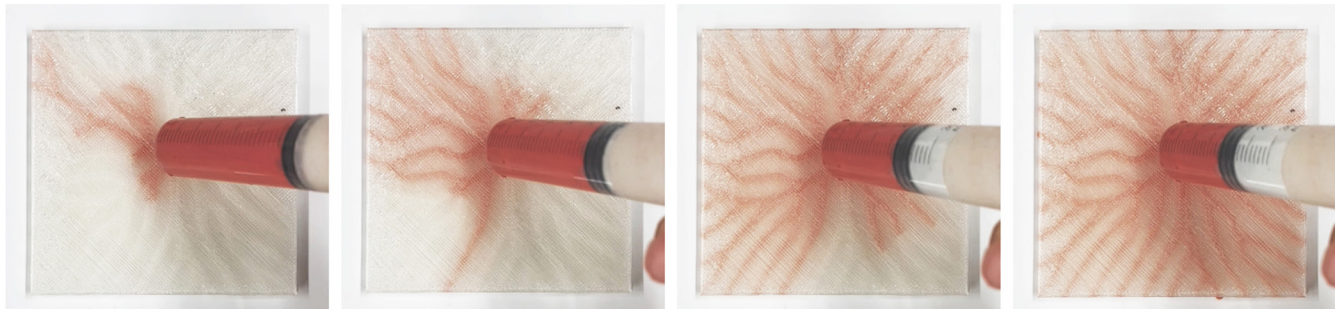
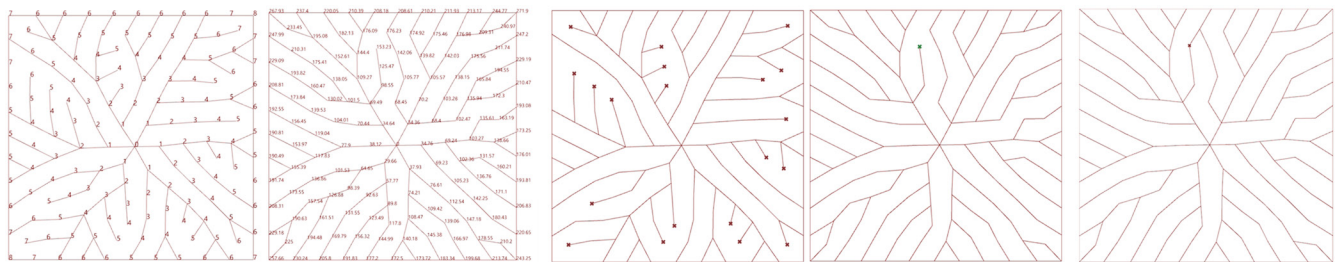
A prototype tile with two inlets and two outlets was designed for connecting multiple tiles into a larger system. The shortest-path algorithm generated four branching patterns from each inlet and outlet and then connected them into a single channel system, extendable by adding more tiles. Fluid flow tests on the prototype and a four-tile system demonstrated the potential for creating a larger interconnected system but also revealed the need for further optimization (Fig. 8).

FABRICATION OF CERAMIC TILES

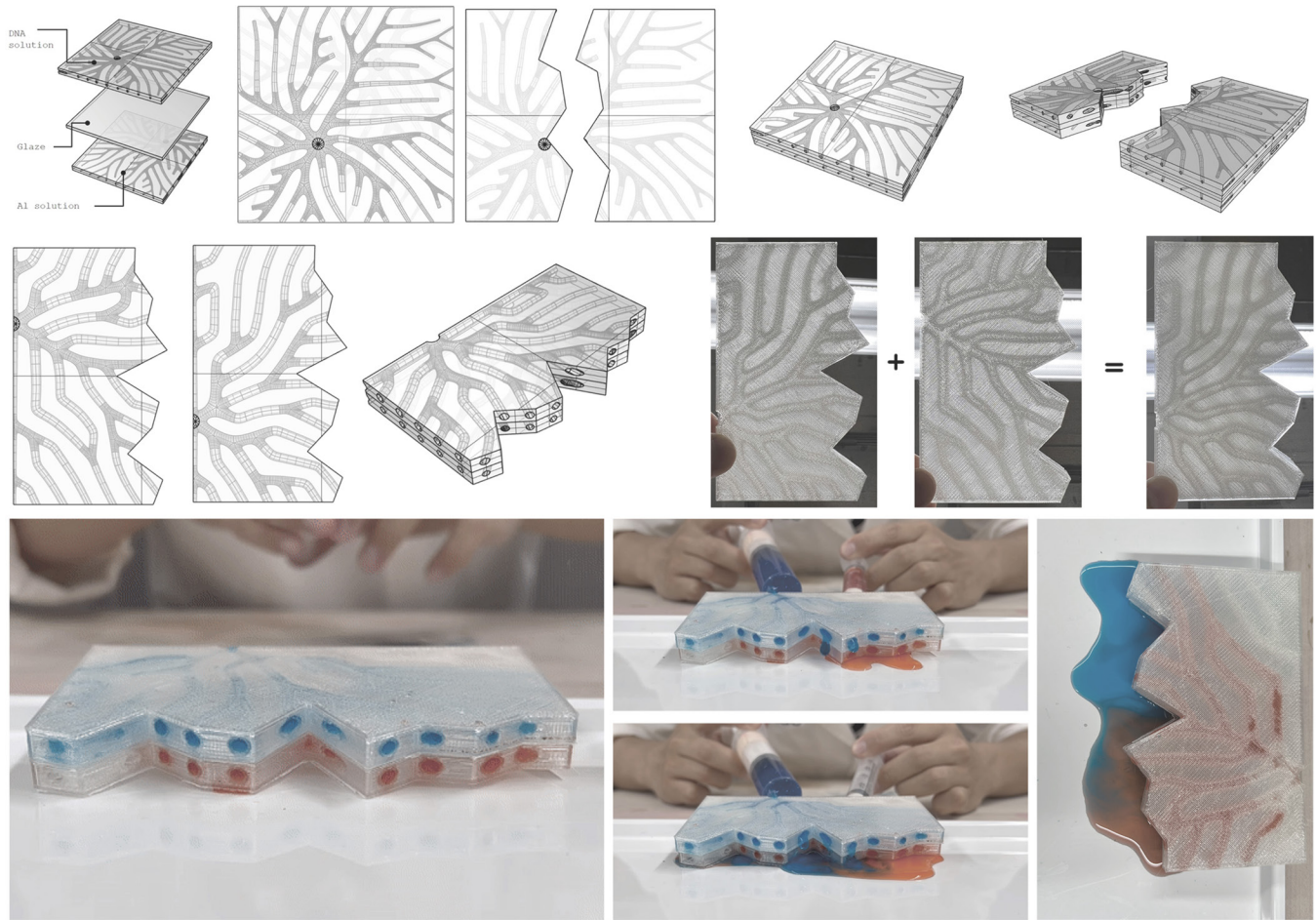
Attempts to use an SLA printer to print tiles with internal channels using ceramic resin were limited by the printer's



5 Initial Tile & Fluid Flow Test



6 Tile with Optimized Channel System & Fluid Flow Test



7 Tiles with Cracks & Fluid Flow Tests

capabilities. The next step involves using the casting method to produce prototype tiles (Fig. 9).

CONCLUSION

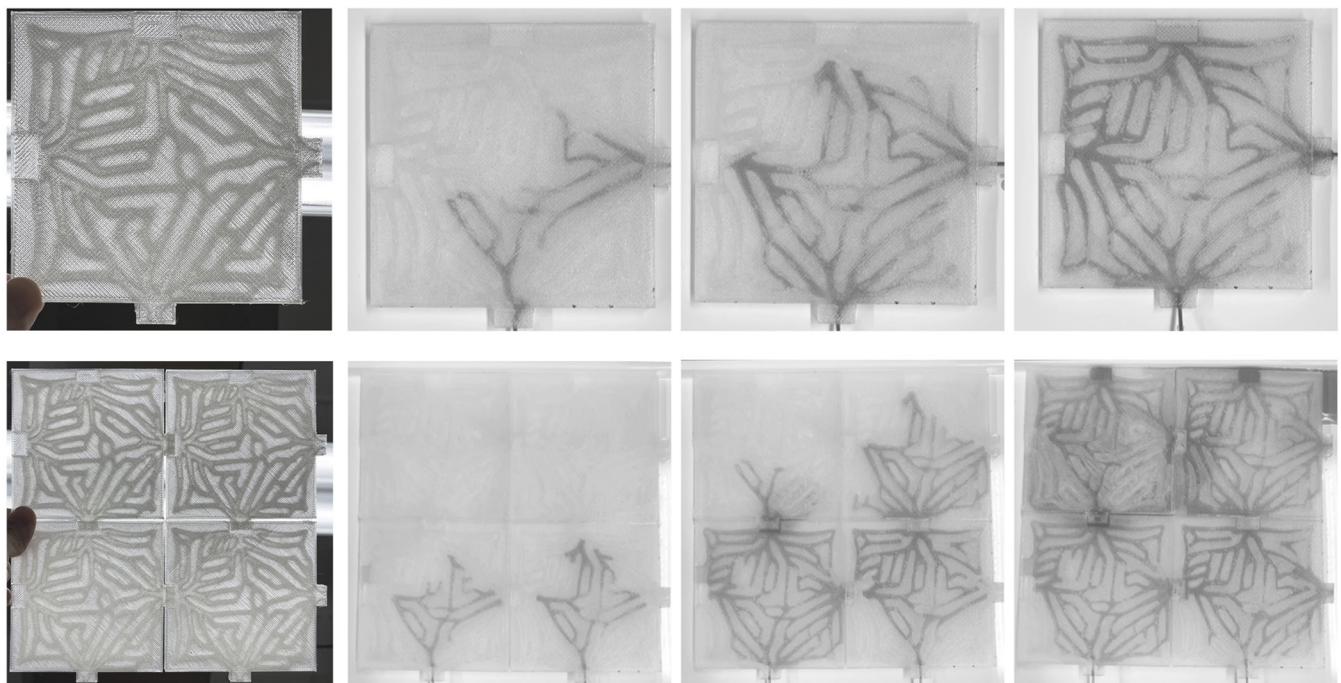
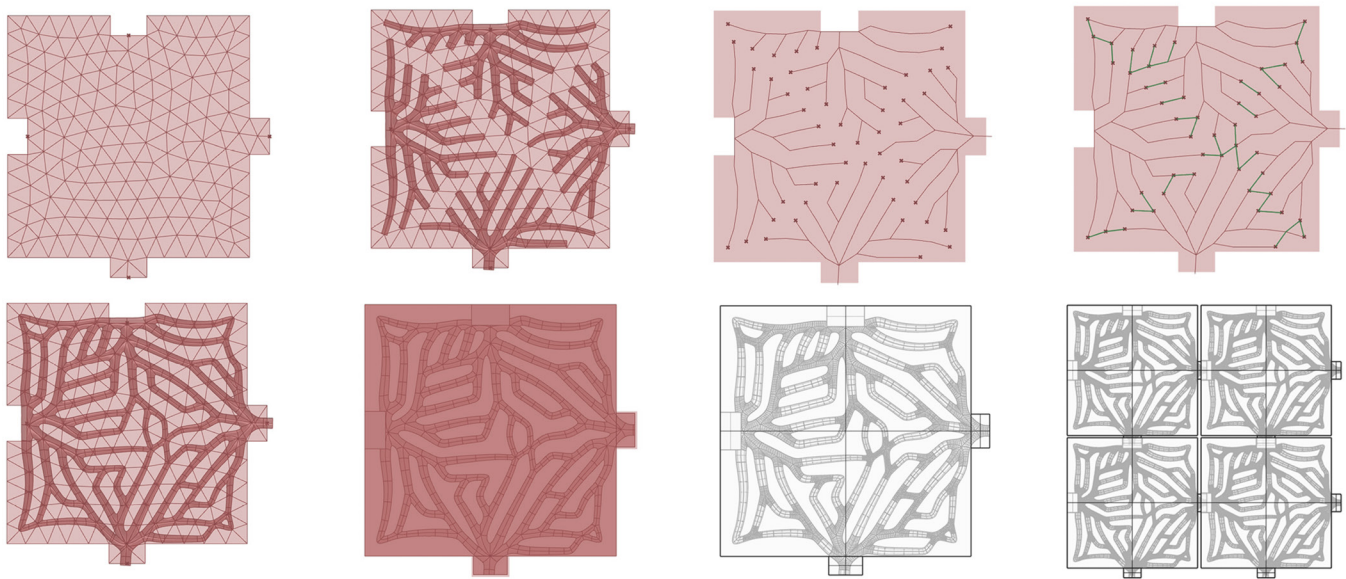
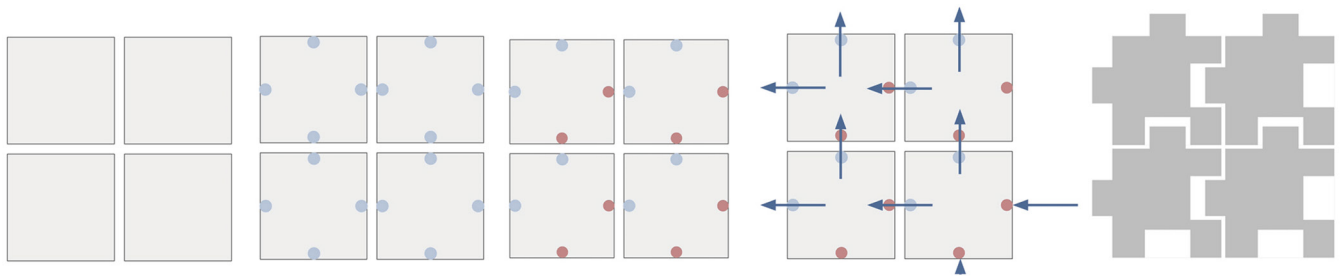
PolyTile 4.0 aims to significantly advance the field of self-healing materials. By integrating DNA hydrogel into ceramic tiles and utilizing computational algorithms to optimize storage channels, this project showcases the potential of advanced computational tools acting as a transformative force in the design field. Future research will extend these findings to other architectural materials, potentially transforming the AEC industries with self-healing capabilities.

ACKNOWLEDGMENTS

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8 Multi-Tile System & Fluid Flow Tests



9 Ceramic Tiles Fabrication Tests

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IMAGE CREDITS

Figure 1: From Blaiszik, B.J., S.L.B. Kramer, S.C. Olugebefola, J.S. Moore, N.R. Sottos, and S.R. White. "Self-Healing Polymers and Composites."

All other drawings and images by the authors.

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Dr. Dan Luo is a professor in Biological and Environmental Engineering at Cornell, who received the NSF CAREER Award, Cornell Provost's Award for Distinguished Scholarship, SUNY Chancellor's Award for Excellence in Scholarship and Creative Activities, NY State Faculty Development Award, CALS Award for Outstanding Accomplishments in Basic Research, and Gates Foundation Point-of-Care Diagnostics Grand Challenge Award. He was selected four times by Cornell's top undergraduates as an outstanding mentor. Dr. Luo has delivered about 300 invited or keynote speeches worldwide. In 2013, he was elected as a College Fellow of the American Institute of Medical and Biological Engineering (AIMBE).

Jenny E. Sabin is the Arthur L. and Isabel B. Wiesenberger Professor in Architecture and the inaugural Chair for the new multicollage Department of Design Tech at the Cornell College of Architecture, Art, and Planning where she established a new advanced research degree in Design Technology. She is principal of Jenny Sabin Studio, an experimental architectural design studio based in Ithaca and Director of the JS Lab at Cornell AAP. Her book, *LabStudio: Design Research Between Architecture and Biology*, co-authored with Peter Lloyd Jones was published in July 2017. In that same year, Sabin won MoMA & MoMA PS1's Young Architects Program with her submission, *Lumen*.